

# Game Theory: Strategic Interaction and Rational Choice

## Part I: Foundations and Rational Choice

Call or Fold?

SciMathSoc

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# Outline

- 1 Introduction
- 2 Actions and Preferences
- 3 Utility and Payoffs
- 4 Theory of Rational Choice
- 5 Games and Strategic Form Representation
- 6 Prisoner's Dilemma
- 7 Dominant and Dominated Strategies
- 8 Iterated Elimination of Dominated Strategies
- 9 Nash Equilibrium
- 10 Star-Delta Method
- 11 Prisoner's Dilemma
- 12 Battle of the Sexes
- 13 Stag Hunt
- 14 Matching Pennies
- 15 Rock Paper Scissors
- 16 Mixed Strategies

# What is Game Theory?

**Game Theory** is the mathematical study of strategic interactions among rational decision-makers.

A situation becomes a game when:

- There are multiple decision makers (players).
- Each player's outcome depends on the actions of others.
- Players act strategically.
- Players seek to maximize their objectives.

Examples:

- Firms competing in a market.
- Political campaigns.
- Auctions.
- Cybersecurity.
- Autonomous AI agents.
- International diplomacy.

# Optimization vs Game Theory

## Optimization

$$\max_{x \in X} u(x)$$

A single decision maker chooses the best action.

## Game Theory

$$u_i(a_1, a_2, \dots, a_n)$$

Each player's payoff depends on actions chosen by all players.

### Key Difference

In optimization, nature is fixed.

In game theory, the environment contains other intelligent agents who react strategically.

# Actions

## Definition

The set of actions available to player  $i$  is denoted by

$$A_i.$$

An action is a feasible decision that a player may choose.

Examples:

| Scenario            | Actions            |
|---------------------|--------------------|
| Prisoner's Dilemma  | Cooperate, Defect  |
| Chess               | Legal chess moves  |
| Market Entry        | Enter, Stay Out    |
| Investment          | Invest, Not Invest |
| Rock-Paper-Scissors | R, P, S            |

# Outcomes and Consequences

When all players choose actions, an outcome is generated.

For two players:

$$(a_1, a_2)$$

For  $n$  players:

$$(a_1, a_2, \dots, a_n)$$

Example:

(Cooperate, Defect)

represents one possible outcome in the Prisoner's Dilemma.

Players ultimately care about outcomes, not merely actions.

# Preferences

Different outcomes are ranked according to desirability.

## Preference Relation

If player  $i$  prefers outcome  $x$  to outcome  $y$ ,

$$x \succ y$$

If player  $i$  is indifferent,

$$x \sim y$$

Weak preference:

$$x \succeq y$$

means that  $x$  is at least as good as  $y$ .

# Properties of Rational Preferences

A rational preference relation usually satisfies:

## 1 **Completeness**

For any outcomes  $x, y$ ,

$$x \succeq y \quad \text{or} \quad y \succeq x$$

## 2 **Transitivity**

If

$$x \succeq y \quad \text{and} \quad y \succeq z,$$

then

$$x \succeq z.$$

These assumptions ensure consistent decision making.



# Utility Functions

Preference rankings can be represented numerically.

## Definition

A utility function for player  $i$  is

$$u_i : X \rightarrow \mathbb{R}$$

where  $X$  denotes the set of outcomes.

Example:

$$u(\text{Pizza}) = 10$$

$$u(\text{Burger}) = 7$$

$$u(\text{Salad}) = 2$$

Higher utility means a more preferred outcome.

# Utility Does Not Measure Happiness

Important interpretation:

Utilities only represent rankings.

Example:

$$u(A) = 100, \quad u(B) = 50$$

does **not** mean outcome  $A$  is twice as enjoyable as outcome  $B$ .

Only the ordering matters:

$$A \succ B$$

Any increasing transformation preserves preferences.

$$u'(x) = 3u(x) + 5$$

represents exactly the same preferences.

# Payoff Functions

In games, utilities depend on actions chosen by all players.

## Definition

$$u_i : A_1 \times A_2 \times \cdots \times A_n \rightarrow \mathbb{R}$$

For player  $i$ ,

$$u_i(a_1, a_2, \dots, a_n)$$

gives the payoff associated with the action profile.

Example:

$$u_1(\text{Enter}, \text{Fight}) = -2$$

means Player 1 receives payoff  $-2$  when entering a market and the incumbent fights.

# Why Payoff Functions Matter

Payoff functions allow us to:

- Represent preferences numerically.
- Compare alternative actions.
- Predict behavior.
- Define equilibrium concepts.
- Analyze strategic interactions mathematically.

Without payoff functions:

- No optimization.
- No best responses.
- No Nash equilibrium analysis.

# Theory of Rational Choice

A rational player chooses the action that maximizes utility.

Formally,

$$a_i^* = \arg \max_{a_i \in A_i} u_i(a_i)$$

Interpretation:

- Evaluate all feasible alternatives.
- Compare their utilities.
- Choose the best one.

This principle forms the foundation of game theory.

# Assumptions Behind Rational Choice

Game theory often assumes:

- 1 Rational players.
- 2 Stable preferences.
- 3 Utility maximization.
- 4 Strategic reasoning.
- 5 Common knowledge of rationality.

## Common Knowledge

Everyone is rational.

Everyone knows everyone is rational.

Everyone knows that everyone knows everyone is rational.

And so on.

# Limitations of Rational Choice

Real humans do not always behave according to perfect rationality.

Examples:

- Bounded rationality.
- Emotional decision making.
- Fairness concerns.
- Reciprocity.
- Altruism.

These observations motivate extensions of classical game theory.

**Next:** Formal definition of games and strategic interaction.

# Definition of a Game

A game is a mathematical model of strategic interaction among rational players.

## Definition

A game is represented by

$$G = (N, \{A_i\}_{i \in N}, \{u_i\}_{i \in N})$$

where

- $N$  = set of players
- $A_i$  = action set of player  $i$
- $u_i$  = payoff function of player  $i$

A game specifies:

- 1 Who plays?
- 2 What can they do?
- 3 What do they receive?



# Elements of a Strategic Situation

Every strategic interaction contains:

- 1 Players
- 2 Actions (Strategies)
- 3 Outcomes
- 4 Preferences
- 5 Payoffs

## Example

Two firms choosing prices:

- Player 1: Firm A
- Player 2: Firm B
- Actions: High Price / Low Price
- Outcome: Market allocation
- Payoffs: Profit

# Normal Form Games

Normal form games represent decisions made simultaneously.

$$G = (N, A_1, \dots, A_n, u_1, \dots, u_n)$$

Characteristics:

- Players choose actions simultaneously.
- No player observes another player's choice before acting.
- Outcomes represented by payoff matrices.

Most classical games are first introduced in normal form.

# Normal Form Representation

For a two-player game:

|     | $L$      | $R$      |
|-----|----------|----------|
| $U$ | $(3, 2)$ | $(1, 4)$ |
| $D$ | $(2, 5)$ | $(0, 1)$ |

Interpretation:

- Rows belong to Player 1.
- Columns belong to Player 2.
- First number = Player 1 payoff.
- Second number = Player 2 payoff.

# Prisoner's Dilemma: Story

Two suspects are arrested.

Each prisoner has two choices:

- ① Cooperate (remain silent)
- ② Defect (betray partner)

They are isolated and cannot communicate.

The prosecutor offers a deal to each prisoner independently.

# Decision Structure

Possible outcomes:

- 1 Both cooperate.
- 2 Prisoner 1 defects, Prisoner 2 cooperates.
- 3 Prisoner 2 defects, Prisoner 1 cooperates.
- 4 Both defect.

Strategic question:

What should a rational prisoner do?

# Prisoner's Dilemma Payoff Matrix

|     | $C$      | $D$      |
|-----|----------|----------|
| $C$ | $(3, 3)$ | $(0, 5)$ |
| $D$ | $(5, 0)$ | $(1, 1)$ |

Interpretation:

- Higher payoff = better outcome.
- Mutual cooperation gives good outcomes.
- Unilateral defection gives highest reward.
- Mutual defection gives poor outcomes.

# Analyzing Prisoner 1

Suppose Prisoner 2 cooperates.

$$3 < 5$$

Therefore Prisoner 1 prefers Defect.

Suppose Prisoner 2 defects.

$$0 < 1$$

Again Prisoner 1 prefers Defect.

Thus:

$$D \succ C$$

regardless of the opponent's action.

# Analyzing Prisoner 2

Suppose Prisoner 1 cooperates.

$$3 < 5$$

Defect is better.

Suppose Prisoner 1 defects.

$$0 < 1$$

Defect remains better.

Therefore Defect is also optimal for Prisoner 2.



# Dominant Strategy

## Definition

A strategy  $s_i$  is dominant if it yields at least as much payoff as every alternative strategy regardless of opponents' actions.

Strict dominance:

$$u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$$

for all  $s_{-i}$ .

In Prisoner's Dilemma: D strictly dominates C.

# Dominated Strategy

## Definition

A strategy is dominated if another strategy performs better against every possible action of opponents.

In Prisoner's Dilemma: C is strictly dominated by D.

Rational players never need to play strictly dominated strategies.

# Strict vs Weak Dominance

## Strict Dominance

$$u_i(s_i, s_{-i}) > u_i(s'_i, s_{-i})$$

for every opponent action.

## Weak Dominance

$$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$$

for every opponent action and

>

for at least one case.

Strict dominance gives stronger conclusions.

# Iterated Elimination of Dominated Strategies (IEDS)

Main idea:

- ➊ Remove dominated strategies.
- ➋ Simplify the game.
- ➌ Re-evaluate remaining strategies.
- ➍ Repeat until no dominated strategies remain.

This process often predicts outcomes before Nash equilibrium analysis.

# Example for IEDS

|     | $L$      | $M$      | $R$      |
|-----|----------|----------|----------|
| $T$ | $(4, 3)$ | $(3, 2)$ | $(2, 1)$ |
| $M$ | $(3, 3)$ | $(2, 2)$ | $(1, 1)$ |
| $B$ | $(1, 4)$ | $(1, 3)$ | $(0, 2)$ |

Observe Player 1's rows.

Compare:

$T$  vs  $M$

# First Elimination

For every column:

$$4 > 3, \quad 3 > 2, \quad 2 > 1$$

Thus,  $T$  strictly dominates  $M$ .

Remove row  $M$ .

Reduced game is as follows:

|     | $L$      | $M$      | $R$      |
|-----|----------|----------|----------|
| $T$ | $(4, 3)$ | $(3, 2)$ | $(2, 1)$ |
| $B$ | $(1, 4)$ | $(1, 3)$ | $(0, 2)$ |

## Second Elimination

Now examine Player 2.

Column  $L$ :

$(3, 4)$

Column  $R$ :

$(1, 2)$

Player 2 prefers  $L$  to  $R$  in every case.

Therefore:

$L$

strictly dominates

$R$

# Final Reduced Game

After elimination:

|     | $L$      | $M$      |
|-----|----------|----------|
| $T$ | $(4, 3)$ | $(3, 2)$ |
| $B$ | $(1, 4)$ | $(1, 3)$ |

Now:  $T$  strictly dominates  $B$ . Predicted outcome:  $(T, L)$  with payoff  $(4, 3)$ .



# Key Takeaways

- A game consists of players, actions and payoff functions.
- Normal form games are represented using payoff matrices.
- Prisoner's Dilemma illustrates strategic conflict.
- Dominant strategies outperform alternatives regardless of opponents.
- Dominated strategies can be eliminated.
- Iterated elimination simplifies strategic analysis.

**Next: Nash Equilibrium and Best Response Analysis**

# Why Do We Need Nash Equilibrium?

Dominant strategies are powerful but rare.

Many games have:

- No dominant strategies
- No dominated strategies
- Strategic dependence

Question:

How can rational players choose actions when their best decision depends on what others do?

Answer:

**Nash Equilibrium**

Players choose mutually consistent strategies.

# Best Response Function

## Definition

A best response of player  $i$  to opponents' actions  $s_{-i}$  is

$$BR_i(s_{-i}) = \arg \max_{s_i} u_i(s_i, s_{-i})$$

Interpretation:

Given what others do, what is my optimal action?

Best responses are the building blocks of Nash equilibrium.

# Example of Best Responses

Consider:

|     | $L$      | $R$      |
|-----|----------|----------|
| $U$ | $(4, 2)$ | $(1, 1)$ |
| $D$ | $(2, 3)$ | $(3, 4)$ |

Player 1:

- If Player 2 chooses  $L$ : choose  $U$
- If Player 2 chooses  $R$ : choose  $D$

Thus,

$$BR_1(L) = U$$

$$BR_1(R) = D$$

# Nash Equilibrium

## Definition

A strategy profile

$$s^* = (s_1^*, s_2^*, \dots, s_n^*)$$

is a Nash Equilibrium if

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$$

for every player  $i$ .

Interpretation:

No player wants to change strategy unilaterally.

At Nash Equilibrium:

“Nobody regrets their decision after seeing everyone else's decision.”

Every player's strategy is a best response to others.

Equivalently:

$$s_i^* \in BR_i(s_{-i}^*)$$

for all players.

# Star-Delta Method

A practical method for locating Nash equilibria.

Procedure:

- 1 Mark Player 1's best responses with a star (\*).
- 2 Mark Player 2's best responses with a delta ( $\Delta$ ).
- 3 Cells containing both symbols are Nash equilibria.

We now apply this method to several classical games.

# Prisoner's Dilemma Revisited

|     | $C$      | $D$      |
|-----|----------|----------|
| $C$ | $(3, 3)$ | $(0, 5)$ |
| $D$ | $(5, 0)$ | $(1, 1)$ |

Best responses:

- Player 1 always prefers  $D$
- Player 2 always prefers  $D$

$(D, D)$

is the unique Nash equilibrium.



# Why Is This Surprising?

Equilibrium:

$$(D, D)$$

Payoffs:

$$(1, 1)$$

Yet both players would prefer

$$(C, C)$$

which gives

$$(3, 3)$$

Nash equilibrium need not maximize total welfare.

# Battle of the Sexes

Two players wish to spend time together.

Choices:

- Opera (O)
- Football (F)

Preferences differ but coordination is valuable.

|          | <i>O</i> | <i>F</i> |
|----------|----------|----------|
| <i>O</i> | (2, 1)   | (0, 0)   |
| <i>F</i> | (0, 0)   | (1, 2)   |

# Best Responses in Battle of the Sexes

|     | $O$      | $F$      |
|-----|----------|----------|
| $O$ | $(2, 1)$ | $(0, 0)$ |
| $F$ | $(0, 0)$ | $(1, 2)$ |

Star-Delta markings:

$(O, O)$

and

$(F, F)$

contain mutual best responses.

Therefore there are two pure Nash equilibria.

# Stag Hunt Game

Two hunters choose:

- Hunt Stag (requires cooperation)
- Hunt Hare (safe option)

|     | $S$      | $H$      |
|-----|----------|----------|
| $S$ | $(4, 4)$ | $(0, 3)$ |
| $H$ | $(3, 0)$ | $(2, 2)$ |

# Equilibria of Stag Hunt

Best-response analysis yields:

$$(S, S)$$

and

$$(H, H)$$

Observations:

- $(S, S)$  is socially efficient.
- $(H, H)$  is safer.
- Coordination is the main challenge.

# Matching Pennies

Player 1 wins if coins match.

Player 2 wins if coins differ.

|     | $H$       | $T$       |
|-----|-----------|-----------|
| $H$ | $(1, -1)$ | $(-1, 1)$ |
| $T$ | $(-1, 1)$ | $(1, -1)$ |

# No Pure Nash Equilibrium

Check every cell:

- Someone always wants to deviate.

Therefore:

No Pure Nash Equilibrium

Motivates mixed strategies.

# Rock Paper Scissors

|     | $R$       | $P$       | $S$       |
|-----|-----------|-----------|-----------|
| $R$ | $(0, 0)$  | $(-1, 1)$ | $(1, -1)$ |
| $P$ | $(1, -1)$ | $(0, 0)$  | $(-1, 1)$ |
| $S$ | $(-1, 1)$ | $(1, -1)$ | $(0, 0)$  |

Every action can be defeated.



# No Pure Equilibrium in RPS

If players choose:

$$(R, R)$$

one player can switch to Paper.

Similarly for every other outcome.

Hence:

No Pure Nash Equilibrium

# Pure vs Mixed Strategies

## Pure Strategy

Choose one action with probability 1.

Example:

$$(R)$$

## Mixed Strategy

Randomize over actions.

$$\sigma_i = (p_1, p_2, \dots, p_k)$$

where

$$\sum p_j = 1$$

# Expected Utility Under Mixed Strategies

Expected payoff:

$$EU_i = \sum_s u_i(s) \Pr(s)$$

Players choose probabilities rather than deterministic actions.

Equilibrium now requires optimal probabilities.

# Mixed Nash Equilibrium

## Principle

At equilibrium, a player must be indifferent among all strategies played with positive probability.

Otherwise they would shift probability toward the better action.

This indifference condition allows us to solve mixed equilibria.

# Matching Pennies: Mixed Equilibrium

Let

$$p = P(H)$$

for Player 1.

Player 2 chooses

$$q = P(H)$$

Indifference condition:

$$q(1) + (1 - q)(-1) = q(-1) + (1 - q)(1)$$

$$q = \frac{1}{2}$$

Similarly:

# Mixed Equilibrium of Matching Pennies

Equilibrium:

$$p = q = \frac{1}{2}$$

Each player randomizes equally.

$$(H, T)$$

are chosen with equal probability.

Expected payoff:

$$0$$

for both players.

# Mixed Equilibrium of Rock-Paper-Scissors

Let

$$p_R, p_P, p_S$$

denote probabilities.

Indifference conditions imply

$$p_R = p_P = p_S$$

and

$$p_R + p_P + p_S = 1$$

Thus

$$p_R = p_P = p_S = \frac{1}{3}$$

# Summary of Nash Equilibrium

- Nash equilibrium is a mutual best-response condition.
- Dominant strategy equilibria are Nash equilibria.
- Games may have:
  - One equilibrium
  - Multiple equilibria
  - No pure equilibrium
- Mixed strategies guarantee equilibrium existence in finite games.

**Next: Extensive Form Games and Sequential Decision Making**



# Why Extensive Form Games?

Normal-form games assume that players move simultaneously.

Many real-world situations are sequential:

- A firm enters a market, then incumbents react.
- A country imposes tariffs, then others respond.
- A company launches a product, competitors adapt.
- An AI agent acts, then other agents observe and react.

To model sequential decision making we use

Extensive Form Games

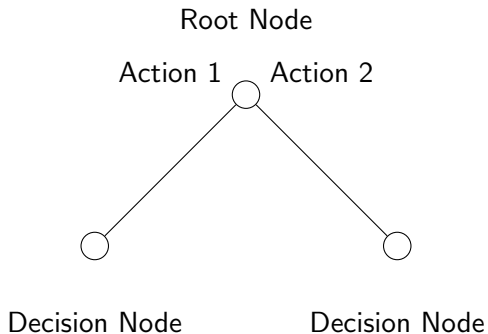
# Extensive Form Representation

An extensive form game consists of:

- 1 Players
- 2 Decision nodes
- 3 Available actions
- 4 Histories
- 5 Information sets
- 6 Payoffs

Games are represented using a game tree.

# Game Tree Components



Key elements:

- Root node
- Decision nodes
- Branches (actions)
- Terminal nodes

# Histories

A history records all actions taken so far.

Examples:

$$h_0 = \emptyset$$

(initial history)

$$h_1 = (\textit{Enter})$$

$$h_2 = (\textit{Enter}, \textit{Fight})$$

The game evolves through histories.

# Action Profiles

Recall:

$$(a_1, a_2, \dots, a_n)$$

denotes an action profile.

In extensive games:

- Different action profiles occur at different stages.
- Later actions may depend on earlier actions.

Thus timing matters.

# Strategies in Extensive Games

A strategy is not merely an action.

A strategy specifies:

What a player will do at every possible decision node.

Therefore a strategy is a complete contingency plan.

Example:

*"If the entrant enters, fight.*

*If the entrant stays out, do nothing."*

# Market Entry Game

Consider two firms:

## **Firm A (Entrant)**

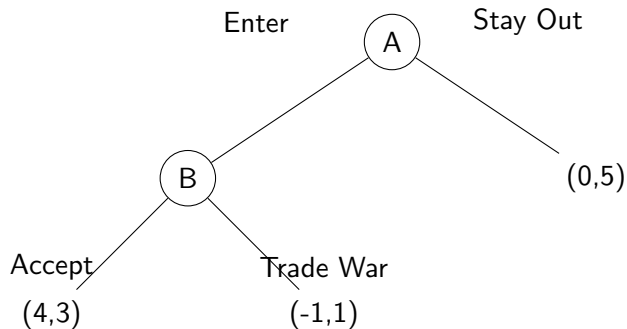
- Enter
- Stay Out

## **Firm B (Incumbent)**

After observing entry:

- Accept competition
- Declare Trade War

# Entry Deterrence Game Tree





# Payoff Interpretation

$$(4, 3)$$

The enterer enters, and the incumbent accepts.

$$(-1, 1)$$

The enterer enters, and the incumbent fights.

$$(0, 5)$$

The entry is excluded.

Question:

What will rational firms do?

# Backward Induction

Key principle:

Analyze the game from the end backward.

Procedure:

- 1 Solve the final decision nodes.
- 2 Replace them with optimal choices.
- 3 Move backward.
- 4 Continue until the root node.

## Step 1: Solve Incumbent's Decision

Suppose that the entry occurs. Firm B chooses between: 3 and 1

Hence, Firm B prefers: Accept

## Step 2: Solve Entrant's Decision

Entrant anticipates acceptance.

Compare:

4

vs

0

Since

$$4 > 0$$

Entrant chooses:

Enter

# Backward Induction Solution

Equilibrium path:

*Enter*  $\rightarrow$  *Accept*

Resulting payoffs:

$(4, 3)$

No player has incentive to deviate.

# Problem with Nash Equilibrium

In sequential games, ordinary Nash equilibrium can allow:

- Non-credible threats
- Irrational future behavior
- Off-path inconsistencies

We need a stronger concept.

# Subgame Perfect Nash Equilibrium (SPNE)

## Definition

A strategy profile is a SPNE if it constitutes a Nash equilibrium in every subgame.

Intuition:

Strategies must remain optimal even after every possible history.

# SPNE in Entry Game

Strategies:

Firm A:

*Enter*

Firm B:

*Accept*

whenever entry occurs.

This strategy profile is optimal in every subgame.

Therefore:

$(\textit{Enter}, \textit{Accept})$



# Trade War Scenario

Country A chooses:

- Open Trade
- Tariff

Country B then chooses:

- Cooperate
- Retaliate

This can be represented as an extensive-form game.

# Trade War Payoffs

Suppose:

$$(Open, Cooperate) = (5, 5)$$

$$(Tariff, Cooperate) = (7, 2)$$

$$(Tariff, Retaliate) = (1, 1)$$

Backward induction predicts whether tariffs are sustainable.

# Beyond Pure Self-Interest

Classical game theory assumes:

$$u_i = \pi_i$$

where  $\pi_i$  denotes material payoff.

However humans often care about others.

Examples:

- Fairness
- Reciprocity
- Charity
- Cooperation

# Altruistic Utility Functions

One possible model:

$$u_i = \pi_i + \alpha\pi_j$$

where

$$\alpha > 0$$

measures concern for others.

Consequences:

- More cooperation
- Greater willingness to contribute
- Better public-good outcomes

# Summary

- Extensive-form games model sequential decisions.
- Strategies are complete contingency plans.
- Game trees represent timing and actions.
- Backward induction solves finite perfect-information games.
- SPNE refines Nash equilibrium.
- Strategic interactions often involve credible threats.
- Human behavior may involve altruism and social preferences.

**Next: Public Goods, Free Riding, and the N-Stag Hunt Game.**

# Social Interactions

Many decisions affect not only ourselves but also others.

Examples:

- Climate change mitigation
- Vaccination
- Open-source software
- National defense
- Public infrastructure

Such situations are known as social interactions.

## Definition

A social dilemma occurs when individually rational actions produce collectively undesirable outcomes.

Characteristics:

- Individual incentives conflict with group welfare.
- Cooperation benefits everyone.
- Defection benefits the individual.

Prisoner's Dilemma is the simplest example.

## **Externality**

A cost or benefit imposed on others that is not reflected in the decision maker's payoff.

Examples:

- Pollution
- Knowledge sharing
- Network effects
- Traffic congestion

Externalities often create social dilemmas.



# Positive and Negative Externalities

## Positive Externalities

- Education
- Scientific research
- Vaccination

## Negative Externalities

- Pollution
- Noise
- Overfishing

Markets may fail when externalities are ignored.

# What is a Public Good?

A public good has two properties:

- 1 Non-rivalrous
- 2 Non-excludable

Examples:

- National defense
- Public parks
- Clean air
- Open-source software
- Scientific knowledge

# Contribution Game

Suppose each player chooses:

$C$  = Contribute

or

$F$  = Free Ride

Contributing costs effort but benefits everyone.

Free-riding avoids the cost while still receiving benefits.

# Two-Player Public Goods Game

|     | $C$      | $F$      |
|-----|----------|----------|
| $C$ | $(4, 4)$ | $(1, 5)$ |
| $F$ | $(5, 1)$ | $(2, 2)$ |

Observations:

- Contribution benefits society.
- Free-riding is individually attractive.

# Free-Rider Problem

Each player reasons:

*"If others contribute, I can enjoy the benefit without paying the cost."*

Result:

$(F, F)$

may emerge despite

$(C, C)$

being socially superior.

This is a classic collective-action problem.

# Tragedy of the Commons

A common resource is shared by many users.

Examples:

- Fisheries
- Groundwater
- Forests
- Grazing land

Each user receives the full benefit of extra usage but bears only part of the cost.

# Mechanism of Collapse

Suppose each user chooses:

Moderate Use

or

Overuse

Short-run incentives encourage overuse.

Collectively:

Resource Depletion

occurs.

Individually rational decisions generate a socially disastrous outcome.

# Self-Interest vs Social Preferences

Classical utility:

$$u_i = \pi_i$$

Behavioral evidence suggests people also care about:

- Fairness
- Reciprocity
- Equality
- Reputation
- Altruism



# Modeling Altruism

Suppose

$$u_i = \pi_i + \alpha \sum_{j \neq i} \pi_j$$

where

$$\alpha > 0$$

represents concern for others.

As  $\alpha$  increases:

- Cooperation becomes more attractive.
- Public-good provision improves.

# Altruism and Cooperation

Consider Prisoner's Dilemma.

Without altruism:

$$(D, D)$$

With sufficient altruism:

$$(C, C)$$

can become individually rational.

Social preferences alter equilibrium behavior.

# From Two Players to Many Players

Most real coordination problems involve many individuals.

Examples:

- Climate agreements
- Team projects
- Open-source communities
- Public infrastructure funding

This motivates the N-player Stag Hunt.

# The N-Stag Hunt

Each player chooses:

$S = \text{Cooperate}$

or

$H = \text{Act Individually}$

The large reward (stag) is obtained only if enough players cooperate.

# Threshold Model

Suppose:

$k$

players are required for success.

If fewer than  $k$  cooperate:

Stag Hunt Fails

If at least  $k$  cooperate:

Large Collective Reward

is achieved.

# Payoff Function

A simple model:

$$u_i = \begin{cases} B - C, & \text{if threshold reached} \\ -C, & \text{if threshold not reached} \end{cases}$$

for contributors.

Non-contributors receive:

$$u_i = \begin{cases} B, & \text{if threshold reached} \\ 0, & \text{otherwise} \end{cases}$$

# Equilibria of N-Stag Hunt

Typical equilibria:

(Everyone Cooperates)

and

(Nobody Cooperates)

The challenge is coordinating expectations.

# Example: Climate Change

Each nation chooses:

- Reduce emissions
- Continue polluting

Benefits of cooperation occur only if many nations participate.

This resembles an N-player Stag Hunt.



# Example: Open Source Software

Developers choose:

- Contribute
- Free Ride

If enough contributors participate:

High-quality software

emerges.

Again, this is a coordination problem.

# Summary of the Entire Lecture

We studied:

- 1 Actions and preferences
- 2 Utility and payoff functions
- 3 Rational choice theory
- 4 Normal-form games
- 5 Prisoner's Dilemma
- 6 Dominance and elimination
- 7 Nash equilibrium
- 8 Mixed strategies
- 9 Extensive-form games
- 10 Backward induction
- 11 Subgame-perfect equilibrium
- 12 Public goods
- 13 Altruism
- 14 N-player coordination

# Key Takeaway

Game Theory provides a mathematical framework for understanding:

Competition

Cooperation

Coordination

Collective Action

From Prisoner's Dilemma to global climate agreements, strategic reasoning helps explain both conflict and cooperation.

Thank You Guys! Apologies for making so many slides in this notes,I'll try to keep it short next time